

THE ELEMENTS

The chart below lays out essential information for each of the known elements. It is arranged by atomic number—the number of protons in the nucleus of an element’s atoms. For each element, the table includes the commonly used

symbol, the relative atomic mass (in this chart given to the nearest two decimal places), and the valency [the number of chemical bonds formed by the atoms] for each given element.

ATOMIC NUMBER	ELEMENT	SYMBOL	ATOMIC MASS	MELTING POINT		BOILING POINT		VALENCY
				°C	°F	°C	°F	
1	Hydrogen	H	1.00	-259	-434	-253	-423	1
2	Helium	He	4.00	-272	-458	-269	-452	0
3	Lithium	Li	6.94	179	354	1,340	2,440	1
4	Beryllium	Be	9.01	1,283	2,341	2,990	5,400	2
5	Boron	B	10.81	2,300	4,170	3,660	6,620	3
6	Carbon	C	12.01	3,500	6,332	4,827	8,721	2,4
7	Nitrogen	N	14.01	-210	-346	-196	-321	3,5
8	Oxygen	O	16.00	-219	-362	-183	-297	2
9	Fluorine	F	19.00	-220	-364	-188	-306	1
10	Neon	Ne	20.18	-249	-416	-246	-410	0
11	Sodium	Na	22.99	98	208	890	1,634	1
12	Magnesium	Mg	24.31	650	1,202	1,105	2,021	2
13	Aluminum	Al	26.98	660	1,220	2,467	4,473	3
14	Silicon	Si	28.09	1,420	2,588	2,355	4,271	4
15	Phosphorus	P	30.97	44	111	280	536	3,5
16	Sulfur	S	32.07	113	235	445	832	2,4,6
17	Chlorine	Cl	35.45	-101	-150	-34	-29	1,3,5,7
18	Argon	Ar	39.95	-189	-308	-186	-303	0
19	Potassium	K	39.10	64	147	754	1,389	1
20	Calcium	Ca	40.08	848	1,558	1,487	2,709	2
21	Scandium	Sc	44.96	1,541	2,806	2,831	5,128	3
22	Titanium	Ti	47.87	1,677	3,051	3,277	5,931	3,4
23	Vanadium	V	50.94	1,917	3,483	3,377	6,111	2,3,4,5
24	Chromium	Cr	52.00	1,903	3,457	2,642	4,788	2,3,6
25	Manganese	Mn	54.94	1,244	2,271	2,041	3,706	2,3,4,6,7
26	Iron	Fe	55.85	1,539	2,802	2,750	4,980	2,3
27	Cobalt	Co	58.93	1,495	2,723	2,877	5,211	2,3
28	Nickel	Ni	58.69	1,455	2,651	2,730	4,950	2,3
29	Copper	Cu	63.55	1,083	1,981	2,582	4,680	1,2
30	Zinc	Zn	65.41	420	788	907	1,665	2
31	Gallium	Ga	69.72	30	86	2,403	4,357	2,3
32	Germanium	Ge	72.63	937	1,719	2,355	4,271	4
33	Arsenic	As	74.92	817	1,503	613	1,135	3,5
34	Selenium	Se	78.96	217	423	685	1,265	2,4,6
35	Bromine	Br	79.90	-7	19	59	138	1,3,5,7
36	Krypton	Kr	83.80	-157	-251	-152	-242	0
37	Rubidium	Rb	85.47	39	102	688	1,270	1
38	Strontium	Sr	87.62	769	1,416	1,384	2,523	2
39	Yttrium	Y	88.91	1,522	2,772	3,338	6,040	3
40	Zirconium	Zr	91.22	1,852	3,366	4,377	7,911	4
41	Niobium	Nb	92.91	2,467	4,473	4,742	8,568	3,5
42	Molybdenum	Mo	95.96	2,610	4,730	5,560	10,040	2,3,4,5,6
43	Technetium	Tc	97.91	2,172	3,942	4,877	8,811	2,3,4,6,7
44	Ruthenium	Ru	101.07	2,310	4,190	3,900	7,052	3,4,6,8
45	Rhodium	Rh	102.91	1,966	3,571	3,727	6,741	3,4
46	Palladium	Pd	106.42	1,554	2,829	2,970	5,378	2,4
47	Silver	Ag	107.87	962	1,764	2,212	4,014	1
48	Cadmium	Cd	112.41	321	610	767	1,413	2
49	Indium	In	114.82	156	313	2,028	3,680	1,3
50	Tin	Sn	118.71	232	450	2,270	4,118	2,4
51	Antimony	Sb	121.76	631	1,168	1,635	2,975	3,5
52	Tellurium	Te	127.60	450	842	990	1,814	2,4,6
53	Iodine	I	126.90	114	237	184	363	1,3,5,7
54	Xenon	Xe	131.29	-112	-170	-107	-161	0
55	Cesium	Cs	132.91	29	84	671	1,240	1
56	Barium	Ba	137.33	725	1,337	1,640	2,984	2
57	Lanthanum	La	138.91	921	1,690	3,457	6,255	3
58	Cerium	Ce	140.12	799	1,470	3,426	6,199	3,4
59	Praseodymium	Pr	140.91	931	1,708	3,512	6,354	3
60	Neodymium	Nd	144.24	1,021	1,870	3,068	5,554	3
61	Promethium	Pm	144.91	1,168	2,134	2,700	4,892	3
62	Samarium	Sm	150.36	1,077	1,971	1,791	3,256	2,3
63	Europium	Eu	151.96	822	1,512	1,597	2,907	2,3

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				°C	°F	°C	°F	
64	Gadolinium	Gd	157.25	1,313	2,395	3,266	5,911	3
65	Terbium	Tb	158.93	1,356	2,473	3,123	5,653	3
66	Dysprosium	Dy	162.50	1,412	2,574	2,562	4,644	3
67	Holmium	Ho	164.93	1,474	2,685	2,695	4,883	3
68	Erbium	Er	167.26	1,529	2,784	2,863	5,185	3
69	Thulium	Tm	168.93	1,545	2,813	1,947	3,537	2,3
70	Ytterbium	Yb	173.04	819	1,506	1,194	2,181	2,3
71	Lutetium	Lu	174.97	1,663	3,025	3,395	6,143	3
72	Hafnium	Hf	178.49	2,227	4,041	4,602	8,316	4
73	Tantalum	Ta	180.95	2,996	5,425	5,427	9,801	3,5
74	Tungsten	W	183.84	3,410	6,170	5,660	10,220	2,4,5,6
75	Rhenium	Re	186.21	3,180	5,756	5,627	10,161	1,4,7
76	Osmium	Os	190.23	3,045	5,510	5,090	9,190	2,3,4,6,8
77	Iridium	Ir	192.22	2,410	4,370	4,130	7,466	3,4
78	Platinum	Pt	195.08	1,772	3,222	3,827	6,921	2,4
79	Gold	Au	196.97	1,064	1,947	2,807	5,080	1,3
80	Mercury	Hg	200.59	-39	-38	357	675	1,2
81	Thallium	Tl	204.38	303	577	1,457	2,655	1,3
82	Lead	Pb	207.20	328	622	1,744	3,171	2,4
83	Bismuth	Bi	208.98	271	520	1,560	2,840	3,5
84	Polonium	Po	208.98	254	489	962	1,764	2,3,4
85	Astatine	At	209.99	300	572	370	698	1,3,5,7
86	Radon	Rn	222.02	-71	-96	-62	-80	0
87	Francium	Fr	223.02	27	81	677	1,251	1
88	Radium	Ra	226.02	700	1,292	1,200	2,190	2
89	Actinium	Ac	227.03	1,050	1,922	3,200	5,792	3
90	Thorium	Th	232.04	1,750	3,182	4,787	8,649	4
91	Protactinium	Pa	231.04	1,597	2,907	4,027	7,281	4,5
92	Uranium	U	238.03	1,132	2,070	3,818	6,904	3,4,5,6
93	Neptunium	Np	237.05	637	1,179	4,090	7,394	2,3,4,5,6
94	Plutonium	Pu	244.06	640	1,184	3,230	5,850	2,3,4,5,6
95	Americium	Am	243.06	994	1,821	2,607	4,724	2,3,4,5,6
96	Curium	Cm	247.07	1,340	2,444	3,190	5,774	2,3,4
97	Berkelium	Bk	247.07	1,050	1,922	710	1,310	2,3,4
98	Californium	Cf	251.08	900	1,652	1,470	2,678	2,3,4
99	Einsteinium	Es	252.08	860	1,580	996	1,825	2,3
100	Fermium	Fm	257.10	unknown		unknown		2,3
101	Mendelevium	Md	258.10	unknown		unknown		2,3
102	Nobelium	No	259.10	unknown		unknown		2,3
103	Lawrencium	Lr	262.11	unknown		unknown		3
104	Rutherfordium	Rf	261.11	unknown		unknown		unknown
105	Dubnium	Db	262.11	unknown		unknown		unknown
106	Seaborgium	Sg	263.12	unknown		unknown		unknown
107	Bohrium	Bh	264.13	unknown		unknown		unknown
108	Hassium	Hs	265.13	unknown		unknown		unknown
109	Meitnerium	Mt	268.14	unknown		unknown		unknown
110	Darmstadtium	Ds	281.16	unknown		unknown		unknown
111	Roentgenium	Rg	273.15	unknown		unknown		unknown
112	Copernicium	Cn	[285]	unknown		unknown		unknown
113	Ununtrium	Uut	[284]	unknown		unknown		unknown
114	Flerovium	Fl	[289]	unknown		unknown		unknown
115	Ununpentium	Uup	[288]	unknown		unknown		unknown
116	Livermorium	Lv	[293]	unknown		unknown		unknown
117	Ununseptium	Uus	[292]	unknown		unknown		unknown
118	Ununoctium	Uuo	[294]	unknown		unknown		unknown